

The Curvature–Packing–Dynamics (CPD) Framework: Paper 3 — The Four Universality Classes in Practice

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This paper presents the four CPD universality classes that arise from the dominant interactions between Curvature (C), Packing (P), Dynamics (Q), and Noise. Rather than providing deep analysis of any specific system, we show that diverse physical, computational, and symbolic systems collapse into a small dictionary of four regimes: C-dominated, P-dominated, Q-dominated, and Noise-dominated. These classes provide a unifying perspective that links toy systems (particle boxes, cellular automata, Chess, Go) with a range of real-world processes. This paper builds directly on CPD Paper 1 (Foundations) and CPD Paper 2 (Predictive Toolkit).

I. RECAP OF CPD (SHORT SUMMARY)

The CPD framework describes any dynamical system through three geometric quantities:

- Curvature C — geometry, spatial form, topological structure,
- Packing P — constraints, compression, bottlenecks, occupancy,
- Dynamics Q — update rules, motion laws, causal propagation.

The Curvature–Compression Number (CCN) provides a simple regime indicator: low for stable geometry-led behavior, high or spiky for constraint-driven or chaotic regimes. Paper 2 introduced CPD as a predictive tool; here we show how systems fall naturally into four universality classes.

II. THE FOUR UNIVERSALITY CLASSES

A. C-Dominated Universality Class (Geometry-led)

Here geometry is the primary driver of evolution. Constraints are mild, and update rules do not override spatial structure. Signs of C-dominance include smooth trajectories, persistent geometric invariants, and low CCN.

Illustrative toy example:

- A particle in a fixed square box: straight-line paths with elastic reflections determined entirely by geometry.

Other systems in this class:

- Circular billiards, harmonic oscillators,
- Two-body gravitational motion,
- Idealized optical reflections in mirror systems.

B. P-Dominated Universality Class (Constraint-led)

Packing, compression, collisions, and bottlenecks dominate behavior. The system’s evolution is driven by forbidden regions and boundary interactions.

Illustrative toy example:

- A particle in an oscillating box: wall motion produces compression spikes, generating bursts in P and CCN.

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Other systems:

- Traffic bottlenecks and jams,
- Granular flows under external forcing,
- Chess midgame, where piece density overrides geometry.

C. Q-Dominated Universality Class (Rule-led)

Update rules dominate system evolution. Geometry acts as a substrate, and packing is secondary. These systems exhibit strongly structured patterns arising from rule interactions.

Illustrative toy examples:

- Wolfram elementary cellular automata (especially Rule 110),
- Chess or Go in modified-rule settings where logical updates overshadow geometry.

Other systems:

- Chemical reaction networks,
- Neural network updates,
- Market microstructure and order-book dynamics.

D. Noise-Dominated Universality Class (Randomness-led)

Randomness overwhelms CPD structure. CCN fluctuates violently; predictions degrade; coherent C, P, Q signatures collapse.

Illustrative toy example:

- Brownian particle under stochastic forcing.

Other systems:

- Financial markets during crises,
- Turbulent diffusion under strong fluctuations,
- Mutation-driven biological processes,
- Noisy communication channels.

III. SUMMARY TABLE OF REPRESENTATIVE SYSTEMS

System	C	P	Q	CCN Regime	Class
Particle in rigid box	high	low	low	stable	C-dominated
Particle in oscillating box	medium	high	low	spikes	P-dominated
Elementary CA (Rule 110)	low	low	high	structured	Q-dominated
Chess/Go complexity	medium	medium	high	mixed	Q-dominated
Brownian particle	low	low	low	chaotic	Noise-dominated

IV. TRANSITION MECHANISMS BETWEEN CLASSES

Systems may shift between universality classes as geometry, constraints, rules, or randomness change. Transitions often reflect simple geometric–dynamic mechanisms rather than domain-specific rules.

Transition Summary Table

Transition	Trigger	Example
$C \rightarrow P$	Geometry perturbation	Oscillating box
$C \rightarrow Q$	Rule dominance	Cellular automaton
$P \rightarrow C$	Constraint relaxation	Traffic clearing
$P \rightarrow Q$	Constraint breakage	Chess exchanges
$Q \rightarrow P$	Rule blocking	CA block formation
$Q \rightarrow \text{Noise}$	Sensitivity spike	Rule 30 chaos
$\text{Any} \rightarrow \text{Noise}$	Strong randomness	Brownian motion

V. WHY UNIVERSALITY CLASSES MATTER

CPD condenses an enormous range of systems into four structural regimes. Once the class is known, prediction strategies become standardized:

- C-led systems use geometric invariants,
- P-led systems use bottleneck and constraint tracking,
- Q-led systems use logical/rule-flow prediction,
- Noise-led systems rely on probabilistic or robustness methods.

This framework converts complexity into geometry-first organization.

VI. ROADMAP TO NEXT PAPERS

- **Paper 4: C-Dominated Systems**
- **Paper 5: P-Dominated Systems**
- **Paper 6: Q-Dominated Systems**
- **Paper 7: Noise-Dominated Systems**

[1] A. Pasternak, *The Curvature–Packing–Dynamics (CPD) Framework: A Geometric Foundation for Structure and Motion*. Zenodo (2025). DOI: 10.5281/zenodo.17717223.